

TERM PROJECT

GROUP 6

User Manual: CO₂ Adsorption in Packed Bed

This MATLAB app simulates the **adsorption of CO₂ gas in a packed bed column** using a one-dimensional ODE model. It helps visualize:

- The gas-phase concentration along the column (C)
- The solid-phase adsorption profile (q)
- Breakthrough curves over time
- Final spatial distributions

The simulation is based on physical and chemical parameters you provide.

Model Basics :

The governing model solves a system of ODEs representing:

- Mass transport (advection-diffusion) of CO₂ in the gas phase
- Rate-limited adsorption into the solid phase
- Axial dispersion
- Nonlinear adsorption kinetics

Numerical integration is done using MATLAB's ode15s.

App Interface Structure :

The app is split into **two main panels**:

Left Panel: Input Parameters

Model Parameters (Optimized)

Column length (m)

1

Spatial points (n)

200

Velocity u (m/s)

0.01

Dispersion D_{ax} (m²/s)

1e-06

Porosity ϵ

0.35

Density ρ_s (kg/m³)

50

Capacity q_{max} (mol/kg)

1

Adsorption k_a (1/s)

0.01

Desorption k_d (1/s)

0.005

Inlet C_0 (mol/m³)

1

Time span (s)

5000

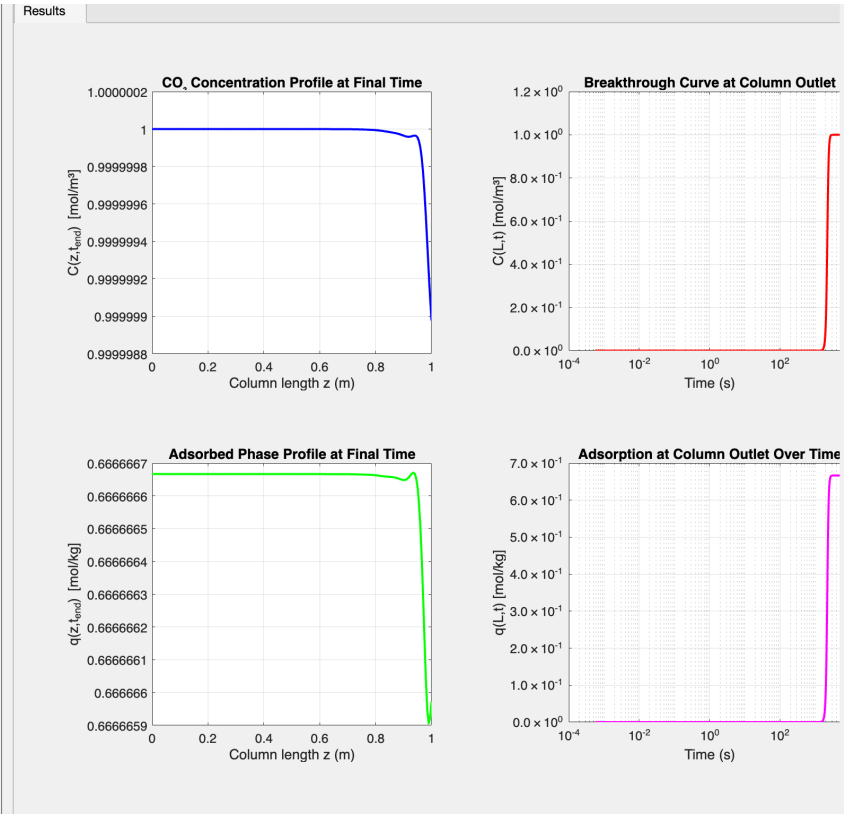
Run Simulation

This panel collects simulation inputs:

Field	Description	Default
Column Length (m)	Length of the adsorption bed	1.0
Spatial Points (n)	Number of discretization points	200
Velocity u (m/s)	Superficial gas velocity	0.01
Dispersion D_{ax} (m ² /s)	Axial dispersion coefficient	1e-6
Porosity ϵ	Bed porosity	0.35
Density ρ_s (kg/m ³)	Solid adsorbent density	50
Capacity q_{max} (mol/kg)	Maximum adsorption capacity	1.0
Adsorption k_a (1/s)	Adsorption rate constant	0.01
Desorption k_d (1/s)	Desorption rate constant	0.005
Inlet C_0 (mol/m ³)	CO ₂ concentration at the inlet	1.0
Time span (s)	Total simulation time	5000

Run Simulation — Press the “Run Simulation” button after setting the inputs.

Right Panel: Results



Divided into **4 subplots** in a 2x2 layout:

- 1. **Top-Left:** Final concentration profile $C(z, t_{end})$
- 2. **Top-Right:** Breakthrough curve $C(L, t)$
- 3. **Bottom-Left:** Final adsorbed phase $q(z, t_{end})$
- 4. **Bottom-Right:** Adsorption evolution at the outlet $q(L, t)$

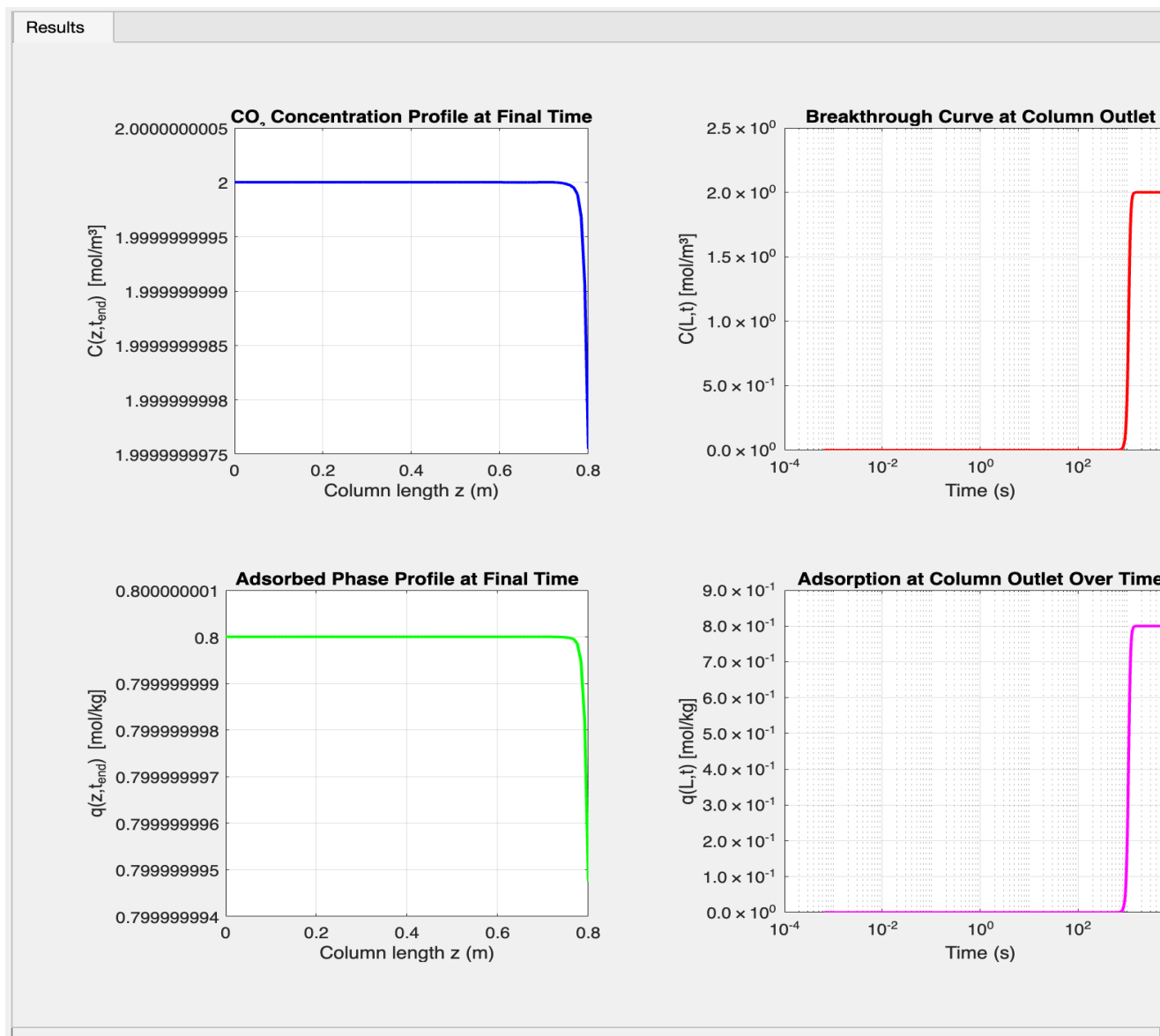
For Examples :

If we take following inputs :

Field	Description	Given Input
Column Length (m)	Length of the adsorption bed	0.8
Spatial Points (n)	Number of discretization points	100
Velocity u (m/s)	Superficial gas velocity	0.01
Dispersion D_{ax} (m ² /s)	Axial dispersion coefficient	1e-6

Field	Description	Given Input
Porosity ε	Bed porosity	0.35
Density ρ_s (kg/m ³)	Solid adsorbent density	50
Capacity $q_{\max}$ (mol/kg)	Maximum adsorption capacity	1.0
Adsorption k_a (1/s)	Adsorption rate constant	0.01
Desorption k_d (1/s)	Desorption rate constant	0.005
Inlet C_0 (mol/m ³)	CO ₂ concentration at the inlet	2.0
Time span (s)	Total simulation time	5000

We Get the Output :



NOTE - In case the app named “Adsorption_Model_App” is not working, you may have to download the “MyAppInstaller_web” file, which I have uploaded.